Koryak Model Summary Report

Prepared from saved model outputs in the project folder. Reaction-time models are listed first, followed by the real brms eye-movement models for LE1 and LE3.

1. Reaction Time Models

Reaction-time models use log speech-onset latency as the outcome. The tables below use Bayesian brms models with the same random-effect structure as the earlier RT analyses. Tables report posterior beta estimates on the log-ms scale, 95% credible intervals, P(beta > 0), and exponentiated ratios.

1.1 All Direct vs All Inverse: Data Summary

Sentence type	n	Mean RT (ms)	SD	Median	Min	Max
direct	496	2573.6	1081.2	2330.5	1029	5981
inverse	383	2645.6	1114.8	2337.0	1105	5981

1.2 All Direct vs All Inverse: Bayesian Contrast

Contrast	Beta log(ms)	95% CI	P(beta > 0)	Ratio	% change	Interpretation
Direct - inverse	-0.040	[-0.106, 0.022]	0.108	0.961	-3.9	direct minus inverse on log RT scale

> Interpretation: the direct-minus-inverse beta is negative, so direct sentences are estimated to be faster. The 95% credible interval still includes zero. P(beta > 0) = 0.108, equivalently P(beta < 0) = 0.892.

1.3 Four-Way RT Model: Data Summary

Condition	n	Mean RT (ms)	SD	Median	Min	Max
direct_1_1	208	2526.1	1052.1	2244.5	1181	5866
direct_1_2	208	2595.7	1091.3	2365.0	1051	5981
inverse_2_2	171	2591.5	1022.7	2310.0	1143	5867
inverse_2_1	196	2699.4	1189.7	2351.5	1105	5903

1.4 Four-Way RT Model: Bayesian Condition Estimates

Condition	Estimated RT (ms)	95% credible interval
direct_1_1	2562	[2195, 2990]
direct_1_2	2639	[2256, 3090]
inverse_2_2	2661	[2274, 3131]
inverse_2_1	2773	[2353, 3283]

1.5 Four-Way RT Model: Bayesian Fixed Effects

Contrast	Beta log(ms)	95% CI	P(beta > 0)	Ratio	% change	Interpretation
Direct 1-2 - direct 1-1	0.030	[-0.071, 0.132]	0.722	1.030	3.0	direct_1_2 minus direct_1_1

Contrast	Beta log(ms)	95% CI	P(beta > 0)	Ratio	% change	Interpretation
Inverse 2-2 - direct 1-1	0.038	[-0.072, 0.145]	0.753	1.038	3.8	inverse_2_2 minus direct_1_1
Inverse 2-1 - direct 1-1	0.079	[-0.027, 0.183]	0.924	1.082	8.2	inverse_2_1 minus direct_1_1

1.6 Four-Way RT Model: Bayesian Pairwise Comparisons

Comparison	Beta log(ms)	95% CI	P(beta > 0)	Ratio A/B	% change
direct_1_1 vs direct_1_2	-0.030	[-0.132, 0.071]	0.278	0.971	-2.9
direct_1_1 vs inverse_2_2	-0.038	[-0.145, 0.072]	0.247	0.963	-3.7
direct_1_1 vs inverse_2_1	-0.079	[-0.183, 0.027]	0.076	0.924	-7.6
direct_1_2 vs inverse_2_2	-0.008	[-0.118, 0.103]	0.442	0.992	-0.8
direct_1_2 vs inverse_2_1	-0.049	[-0.162, 0.058]	0.182	0.952	-4.8
inverse_2_2 vs inverse_2_1	-0.041	[-0.160, 0.074]	0.244	0.960	-4.0

1.7 Bayesian RT Model Diagnostics

Chains	Iter	Warmup	adapt_delta	max_treedepth
4	2000	1000	0.99	12

Model	Term	Rhat	Bulk ESS	Tail ESS
direct_inverse	Intercept	1.0079	908	1546
direct_inverse	direct	1.0008	3327	3151
fourway	Intercept	1.0045	791	1246
fourway	condition4direct_1_2	1.0005	2247	2545
fourway	condition4inverse_2_2	1.0004	2402	2624
fourway	condition4inverse_2_1	1.0006	2119	2610

Model	Max Rhat	Min bulk ESS	Min tail ESS	Worst bulk variable	Worst tail variable
direct_inverse	1.0105	173	413	sd_item__direct	sd_item__direct
fourway	1.0116	372	725	sd_item__condition4inverse_2_2	sd_item__condition4inverse_2_1

> Diagnostics: both Bayesian RT models had no divergent transitions and no max-treedepth hits. Fixed-effect diagnostics were good, with Rhat approximately 1.00 and fixed-effect bulk ESS at least 791. Stan warned about low bulk ESS for nuisance random-effect SD parameters; this is reported in the all-parameter table.

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2. Bayesian Eye-Movement Models

Bayesian models are brms Gaussian models over empirical logit-transformed fixation proportions, fit separately for agent looks and patient looks. Time is modeled with orthogonal polynomial terms up to cubic. All contrast-coded condition terms use +1/-1 coding, so the model beta is half of the full condition difference; the report also gives 2*beta for contrast and contrast-by-time terms.

2.1 brms Run Settings

Model set	Settings
LE1 all direct vs inverse	4 chains, 2000 iter, 1000 warmup, adapt_delta .99, max_treedepth 12
LE1 pairwise	4 chains, 2000 iter, 1000 warmup, adapt_delta .99, max_treedepth 12
LE3 all direct vs inverse	4 chains, 2000 iter, 1000 warmup, adapt_delta .99, max_treedepth 12
LE3 pairwise	4 chains, 2000 iter, 1000 warmup, adapt_delta .99, max_treedepth 12

2.2 Fixed-Effect Convergence Diagnostics

Window	Model set	Max Rhat	Min bulk ESS	Min tail ESS
LE1	all direct vs all inverse	1.0046	966	1685
LE1	number pairwise	1.0055	1014	1579
LE3	all direct vs all inverse	1.0046	1169	1816
LE3	number pairwise	1.0097	807	1301

2.3 All-Parameter Convergence Diagnostics

Window	Model set	Comparison	Looks	Max Rhat	Min bulk ESS	Min tail ESS	Worst bulk variable	Worst tail variable
LE1	All direct vs all inverse	direct inverse	agent looks	1.0050	735	667	lp__	sd_participant__time
LE1	All direct vs all inverse	direct inverse	patient looks	1.0147	603	1190	sd_participant__time	sd_participant__Intercept
LE1	Number pairwise	direct 1 1 vs direct 1 2	agent looks	1.0191	480	1567	sd_participant__Intercept	lp__
LE1	Number pairwise	direct 1 1 vs direct 1 2	patient looks	1.0156	448	824	sd_participant__Intercept	sd_participant__Intercept
LE1	Number pairwise	direct 1 1 vs inverse 2 2	agent looks	1.0076	794	991	sd_participant__time	sd_participant__Intercept
LE1	Number pairwise	direct 1 1 vs inverse 2 2	patient looks	1.0103	636	747	sd_participant__time	sd_participant__Intercept
LE1	Number pairwise	inverse 2 1 vs inverse 2 2	agent looks	1.0060	615	1546	sd_participant__Intercept	lp__
LE1	Number pairwise	inverse 2 1 vs inverse 2 2	patient looks	1.0077	433	1313	sd_participant__Intercept	sd_participant__Intercept
LE3	All direct vs all inverse	direct inverse	agent looks	1.0066	657	692	lp__	sd_participant__time
LE3	All direct vs all inverse	direct inverse	patient looks	1.0051	736	325	lp__	r_participant_condition[K10.direct,Intercept]
LE3	Number pairwise	direct 1 1 vs direct 1 2	agent looks	1.0111	770	1144	lp__	sd_participant__time
LE3	Number pairwise	direct 1 1 vs direct 1 2	patient looks	1.0069	590	980	sd_participant__time	sd_participant__Intercept
LE3	Number pairwise	direct 1 1 vs inverse 2 2	agent looks	1.0054	759	651	sd_participant__time	sd_participant__Intercept
LE3	Number pairwise	direct 1 1 vs inverse 2 2	patient looks	1.0078	511	845	sd_participant__Intercept	sd_participant__Intercept
LE3	Number pairwise	inverse 2 1 vs inverse 2 2	agent looks	1.0134	455	522	sd_participant__Intercept	sd_participant__Intercept
LE3	Number pairwise	inverse 2 1 vs inverse 2 2	patient looks	1.0073	580	1278	sd_participant__time	sd_participant__Intercept

> Log check: no divergence or treedepth warnings were found. The LE3 all-direct-vs-all-inverse run reported one low tail-ESS warning; this is reflected in the all-parameter diagnostics above.

2.4 LE1 all direct vs all inverse: Fixed Effects

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
all direct vs all inverse	agent looks	Intercept	0.796	[0.344, 1.266]	0.999	

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
all direct vs all inverse	agent looks	Time polynomial 1	27.691	[8.470, 47.711]	0.997	
all direct vs all inverse	agent looks	Time polynomial 2	-2.540	[-7.875, 2.555]	0.174	
all direct vs all inverse	agent looks	Time polynomial 3	-4.108	[-9.077, 0.896]	0.051	
all direct vs all inverse	agent looks	direct(+1) vs inverse(-1)	0.069	[-0.138, 0.285]	0.753	0.138 [-0.275, 0.570]
all direct vs all inverse	agent looks	Time polynomial 1 x direct(+1) vs inverse(-1)	12.966	[0.895, 25.891]	0.982	25.933 [1.791, 51.782]
all direct vs all inverse	agent looks	Time polynomial 2 x direct(+1) vs inverse(-1)	4.054	[-1.084, 9.247]	0.932	8.109 [-2.167, 18.493]
all direct vs all inverse	agent looks	Time polynomial 3 x direct(+1) vs inverse(-1)	-1.166	[-6.463, 4.077]	0.327	-2.332 [-12.925, 8.155]
all direct vs all inverse	patient looks	Intercept	-1.818	[-2.491, -1.161]	0.000	
all direct vs all inverse	patient looks	Time polynomial 1	-54.628	[-78.268, -31.273]	0.000	
all direct vs all inverse	patient looks	Time polynomial 2	10.108	[4.152, 15.958]	1.000	
all direct vs all inverse	patient looks	Time polynomial 3	-2.107	[-8.462, 4.332]	0.247	
all direct vs all inverse	patient looks	direct(+1) vs inverse(-1)	0.002	[-0.327, 0.343]	0.503	0.004 [-0.654, 0.685]
all direct vs all inverse	patient looks	Time polynomial 1 x direct(+1) vs inverse(-1)	-7.607	[-26.280, 10.851]	0.211	-15.215 [-52.561, 21.702]
all direct vs all inverse	patient looks	Time polynomial 2 x direct(+1) vs inverse(-1)	-3.730	[-9.575, 2.271]	0.108	-7.459 [-19.150, 4.541]
all direct vs all inverse	patient looks	Time polynomial 3 x direct(+1) vs inverse(-1)	3.429	[-2.998, 9.723]	0.858	6.857 [-5.995, 19.447]

2.4 LE1 pairwise contrasts: Fixed Effects

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Intercept	0.773	[0.222, 1.310]	0.996	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 1	23.974	[6.821, 41.686]	0.996	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 2	3.774	[-1.324, 8.960]	0.920	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 3	-2.575	[-7.750, 2.449]	0.161	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	condition contrast(+1/-1)	0.089	[-0.359, 0.530]	0.659	0.178 [-0.718, 1.059]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	0.861	[-14.118, 15.670]	0.542	1.722 [-28.235, 31.340]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	-0.776	[-5.788, 4.304]	0.382	-1.552 [-11.576, 8.608]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	2.571	[-2.154, 7.570]	0.851	5.141 [-4.308, 15.141]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Intercept	-1.928	[-2.632, -1.231]	0.000	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 1	-38.463	[-59.030, -18.218]	2.50e-04	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 2	3.143	[-2.482, 9.083]	0.849	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 3	-0.022	[-5.821, 5.715]	0.496	

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	condition contrast(+1/-1)	-0.242	[-0.799, 0.308]	0.199	-0.484 [-1.597, 0.616]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	-6.216	[-23.954, 10.685]	0.234	-12.431 [-47.907, 21.371]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	7.162	[1.339, 13.232]	0.993	14.324 [2.678, 26.464]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	-9.030	[-15.167, -3.081]	0.001	-18.059 [-30.335, -6.161]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Intercept	0.823	[0.250, 1.412]	0.997	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1	10.997	[-7.020, 28.652]	0.888	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2	-6.590	[-11.838, -1.443]	0.008	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3	-0.491	[-5.554, 4.703]	0.429	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	condition contrast(+1/-1)	0.040	[-0.421, 0.504]	0.575	0.081 [-0.842, 1.008]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	-7.389	[-22.953, 8.421]	0.175	-14.778 [-45.907, 16.841]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	7.390	[2.271, 12.344]	0.999	14.779 [4.541, 24.689]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	-5.820	[-10.803, -0.576]	0.014	-11.640 [-21.606, -1.151]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Intercept	-1.844	[-2.662, -1.069]	0.001	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1	-29.752	[-53.298, -7.487]	0.007	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2	11.128	[4.752, 17.340]	1.000	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3	-4.415	[-10.667, 1.738]	0.076	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	condition contrast(+1/-1)	-0.028	[-0.725, 0.669]	0.467	-0.055 [-1.451, 1.338]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	7.169	[-13.916, 27.947]	0.755	14.339 [-27.832, 55.895]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	-23.829	[-30.056, -17.911]	0.000	-47.657 [-60.112, -35.823]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	0.897	[-5.349, 7.136]	0.606	1.793 [-10.698, 14.271]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Intercept	0.836	[0.223, 1.464]	0.996	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1	20.583	[1.439, 40.686]	0.982	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2	-5.368	[-10.562, -0.007]	0.025	

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3	2.741	[-2.661, 8.201]	0.839	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	condition contrast(+1/-1)	0.054	[-0.339, 0.460]	0.606	0.109 [-0.678, 0.919]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	2.995	[-10.831, 17.002]	0.671	5.990 [-21.662, 34.004]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	8.229	[2.924, 13.631]	0.998	16.458 [5.849, 27.261]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	-2.678	[-7.884, 2.673]	0.157	-5.357 [-15.768, 5.347]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Intercept	-1.997	[-2.734, -1.280]	0.000	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1	-40.290	[-64.292, -17.467]	0.001	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2	21.945	[15.947, 27.993]	1.000	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3	-6.705	[-12.564, -0.801]	0.013	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	condition contrast(+1/-1)	-0.167	[-0.693, 0.349]	0.259	-0.334 [-1.386, 0.698]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	-3.005	[-21.249, 15.127]	0.371	-6.011 [-42.499, 30.255]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	-12.435	[-18.405, -6.630]	0.000	-24.869 [-36.810, -13.260]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	-1.708	[-7.611, 4.295]	0.293	-3.416 [-15.222, 8.591]

2.4 LE3 all direct vs all inverse: Fixed Effects

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
all direct vs all inverse	agent looks	Intercept	0.032	[-0.617, 0.679]	0.545	
all direct vs all inverse	agent looks	Time polynomial 1	-50.957	[-73.662, -28.770]	0.000	
all direct vs all inverse	agent looks	Time polynomial 2	-3.950	[-9.582, 1.679]	0.084	
all direct vs all inverse	agent looks	Time polynomial 3	-2.447	[-8.035, 3.293]	0.191	
all direct vs all inverse	agent looks	direct(+1) vs inverse(-1)	-0.313	[-0.559, -0.071]	0.009	-0.627 [-1.118, -0.141]
all direct vs all inverse	agent looks	Time polynomial 1 x direct(+1) vs inverse(-1)	-13.756	[-24.039, -4.424]	0.003	-27.512 [-48.078, -8.847]
all direct vs all inverse	agent looks	Time polynomial 2 x direct(+1) vs inverse(-1)	5.305	[-0.217, 10.637]	0.972	10.610 [-0.434, 21.273]
all direct vs all inverse	agent looks	Time polynomial 3 x direct(+1) vs inverse(-1)	-4.401	[-9.848, 1.003]	0.062	-8.802 [-19.697, 2.007]
all direct vs all inverse	patient looks	Intercept	-0.908	[-1.484, -0.336]	0.002	
all direct vs all inverse	patient looks	Time polynomial 1	47.548	[31.501, 64.704]	1.000	
all direct vs all inverse	patient looks	Time polynomial 2	-5.493	[-11.282, 0.541]	0.037	
all direct vs all inverse	patient looks	Time polynomial 3	15.052	[9.641, 20.500]	1.000	
all direct vs all inverse	patient looks	direct(+1) vs inverse(-1)	0.327	[0.042, 0.613]	0.986	0.654 [0.085, 1.226]
all direct vs all inverse	patient looks	Time polynomial 1 x direct(+1) vs inverse(-1)	1.394	[-11.770, 15.022]	0.583	2.787 [-23.539, 30.043]

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
all direct vs all inverse	patient looks	Time polynomial 2 x direct(+1) vs inverse(-1)	1.054	[-4.666, 6.849]	0.647	2.109 [-9.332, 13.699]
all direct vs all inverse	patient looks	Time polynomial 3 x direct(+1) vs inverse(-1)	-1.096	[-6.804, 4.669]	0.350	-2.191 [-13.608, 9.338]

2.4 LE3 pairwise contrasts: Fixed Effects

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Intercept	-0.476	[-1.409, 0.423]	0.147	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 1	-33.945	[-56.179, -11.772]	0.003	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 2	2.644	[-2.964, 8.252]	0.817	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 3	-4.868	[-10.483, 0.868]	0.047	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	condition contrast(+1/-1)	-0.098	[-0.640, 0.431]	0.366	-0.197 [-1.280, 0.861]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	3.163	[-12.767, 18.857]	0.664	6.325 [-25.535, 37.715]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	1.344	[-4.471, 6.933]	0.675	2.688 [-8.943, 13.866]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	-4.395	[-10.199, 1.582]	0.069	-8.790 [-20.397, 3.163]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Intercept	-0.506	[-1.322, 0.278]	0.106	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 1	28.975	[12.381, 45.458]	1.000	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 2	-2.500	[-8.065, 2.920]	0.180	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 3	11.100	[5.774, 16.234]	1.000	
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	condition contrast(+1/-1)	-0.019	[-0.563, 0.553]	0.470	-0.039 [-1.126, 1.105]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	-5.945	[-20.638, 8.536]	0.204	-11.891 [-41.277, 17.072]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	-5.624	[-11.004, -0.184]	0.021	-11.248 [-22.008, -0.367]
direct 1 agent, 1 patient vs direct 1 agent, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	5.481	[0.251, 10.621]	0.980	10.962 [0.501, 21.242]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Intercept	0.588	[-0.080, 1.278]	0.960	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1	-27.123	[-44.088, -9.676]	0.002	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2	-6.956	[-12.311, -1.805]	0.005	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3	1.034	[-4.192, 6.454]	0.650	

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	condition contrast(+1/-1)	0.209	[-0.256, 0.688]	0.805	0.418 [-0.512, 1.376]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	-0.669	[-15.326, 15.013]	0.462	-1.338 [-30.653, 30.027]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	-5.997	[-11.411, -0.508]	0.015	-11.993 [-22.822, -1.016]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	-0.693	[-5.810, 4.471]	0.398	-1.385 [-11.620, 8.942]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Intercept	-1.536	[-2.313, -0.754]	0.001	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1	34.024	[11.461, 56.660]	0.998	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2	-3.964	[-9.574, 1.673]	0.091	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3	11.814	[6.041, 17.618]	1.000	
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	condition contrast(+1/-1)	-0.264	[-0.853, 0.365]	0.192	-0.527 [-1.706, 0.730]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	-3.552	[-22.510, 14.568]	0.358	-7.103 [-45.019, 29.136]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	-3.094	[-8.700, 2.758]	0.142	-6.188 [-17.400, 5.515]
inverse 2 agents, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	1.200	[-4.693, 6.885]	0.659	2.399 [-9.387, 13.769]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Intercept	-0.142	[-0.845, 0.550]	0.352	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1	-28.376	[-45.565, -12.023]	0.001	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2	1.314	[-4.206, 6.848]	0.693	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3	-3.573	[-8.785, 1.612]	0.083	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	condition contrast(+1/-1)	-0.421	[-0.936, 0.104]	0.056	-0.842 [-1.872, 0.208]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 1 x condition contrast(+1/-1)	-0.970	[-14.422, 13.181]	0.440	-1.941 [-28.845, 26.361]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 2 x condition contrast(+1/-1)	2.333	[-3.115, 7.864]	0.803	4.665 [-6.231, 15.727]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	agent looks	Time polynomial 3 x condition contrast(+1/-1)	-5.327	[-10.759, 0.247]	0.030	-10.654 [-21.518, 0.493]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Intercept	-0.893	[-1.664, -0.162]	0.010	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1	29.212	[12.629, 46.327]	0.999	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2	-4.336	[-9.787, 0.816]	0.049	

Comparison	Looks	Term	Beta	95% CI	P(beta > 0)	Full difference 2*beta
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3	13.259	[7.828, 18.827]	1.000	
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	condition contrast(+1/-1)	0.379	[-0.224, 0.977]	0.899	0.758 [-0.449, 1.955]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 1 x condition contrast(+1/-1)	-6.720	[-22.205, 8.348]	0.192	-13.439 [-44.410, 16.696]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 2 x condition contrast(+1/-1)	-3.588	[-8.861, 1.483]	0.085	-7.177 [-17.721, 2.966]
direct 1 agent, 1 patient vs inverse 2 agents, 2 patients	patient looks	Time polynomial 3 x condition contrast(+1/-1)	2.580	[-2.719, 7.786]	0.840	5.160 [-5.439, 15.572]